综合练习

人员

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上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/4164

比赛概况 题目列表 选择题列表 提交记录 实时榜单 选择题排行榜 编辑比赛

王向东老师周日十点半C++树状数组

刷新

#	用户名	姓名	编程分	时间	A	B	C	D
1	yangjinshuo	杨谨硕	300	509	100	100	100	
2	zhouziyi	周子一	300	1779	100	100	100	
3	chenluoran	陈洛冉	300	3178	100	100	100	
4	lizihan	李子瀚	200	969	100	100		
5	yuzijia1	于子珈	200	1001	100	100		0
6	qinxiansen	秦显森	200	3087	100	100	0	

本周作业

https://cppoj.kids123code.com/contest/4304 (课上讲了 A ~ B 题, 课后作业是 C 题必做)

课堂表现

同学们课上做题整体都不太熟, 还是对树状数组和线段树不熟, 需要课下加强这两个的练习。

课堂内容

贪婪大陆

两个树状数组维护, 一个维护 l, 一个维护 r

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1e5 + 5;
int tr_1[maxn], tr_2[maxn];

int lowbit(int x) { return x&(-x); }
void update(int tr[], int x, int k) {
  while (x < maxn) tr[x] += k, x += lowbit(x);
}
int query(int tr[], int x) {
  int res = 0;
```

```
    while (x) res += tr[x], x -= lowbit(x);
    return res;
}

int main()
{
    int n, m; cin >> n >> m;
    while (m -- ) {
        int op, l, r; cin >> op >> l >> r;
        if (op == 1) update(tr_1, l, 1), update(tr_2, r, 1);
        else cout << query(tr_1, r) - query(tr_2, l-1) << endl;
    }
    return 0;
}
```

扶苏的问题

维护 2 个懒标签, flag 和 add

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 1e6 + 5;
const int inf = 0x3f3f3f3f3f3f3f3f;
struct node {
    int l, r, maxx, flag, add;
} tr[maxn*4];
int w[maxn];

void pushup(int u) { tr[u].maxx = max(tr[u*2].maxx, tr[u*2+1].maxx); }
void pushdown(int u) {
    if (tr[u].flag != inf) {
        int v = tr[u].flag + tr[u].add;
        tr[u*2].flag = v, tr[u*2].add = 0, tr[u*2].maxx = v;
        tr[u*2+1].flag = v, tr[u*2+1].add = 0, tr[u*2+1].maxx = v;
        tr[u].flag = inf, tr[u].add = 0;
    } else if (tr[u].add) {
        tr[u*2].add += tr[u].add, tr[u*2].maxx += tr[u].add;
        tr[u*2+1].add += tr[u].add, tr[u*2+1].maxx += tr[u].add;
        tr[u].add = 0;
    }
}
void build(int u, int l, int r) {
    tr[u] = {l, r, 0, inf, 0};
    if (l == r) { tr[u].maxx = w[l]; return; }

    int mid = (l + r) / 2;
    build(u*2, l, mid), build(u*2+1, mid+1, r);
    pushup(u);
```

```
}
void modify(int u, int l, int r, bool is_flag, int k) {
  if (tr[u].l>=l && tr[u].r<=r) {
    if (is_flag) tr[u].flag = k, tr[u].add = 0, tr[u].maxx = k;
    else tr[u].add += k, tr[u].maxx += k;
    return;
  }

  pushdown(u);
  int mid = (tr[u].l + tr[u].r) / 2;
  if (l <= mid) modify(u*2, l, r, is_flag, k);
  if (r > mid) modify(u*2+1, l, r, is_flag, k);
  pushup(u);
}
int query(int u, int l, int r) {
  if (tr[u].l>=l && tr[u].r<=r) return tr[u].maxx;

  pushdown(u);
  int mid = (tr[u].l + tr[u].r) / 2, res = -inf;
  if (l <= mid) res = max(res, query(u*2, l, r));
  if (r > mid) res = max(res, query(u*2+1, l, r));
  return res;
}

signed main()
{
  ios::sync_with_stdio(false);
  cin.tie(0); cout.tie(0);

  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) cin >> w[i];
  build(1, 1, n);

  while (m -- ) {
    int op, l, r; cin >> op >> l >> r;
    if (op == 1) {
      int k; cin >> k; modify(1, l, r, true, k);
    } else if (op == 2) {
      int k; cin >> k; modify(1, l, r, false, k);
    } else cout << query(1, l, r) << "\n";
  }
  return 0;
}
```

Transportation

根据 机场 和 港口 是否使用, 分为 4 次最小生成树来做

```
#include <bits/stdc++.h>
#define int long long

```

```
using namespace std;

const int maxn = 2e5 + 5;
struct node {
  int from, to, value;
  bool operator < (const node& p) const { return value < p.value; }
};
vector<node> vec;
int X[maxn], Y[maxn], A[maxn], B[maxn], Z[maxn];
int n, m;

int f[maxn];
int fFind(int x) {
  if (f[x] != x) f[x] = fFind(f[x]);
  return f[x];
}

int kruskal(int flag1, int flag2) {
  for (int i = 1; i <= n+2; ++i) f[i] = i;
  int res = 0, cnt = n-1 + flag1 + flag2;
  for (node it : vec) {
    int a = it.from, b = it.to, c = it.value;
    if (!flag1 && (a==n+1||b==n+1)) continue;
    if (!flag2 && (a==n+2||b==n+2)) continue;
    int fa = fFind(a), fb = fFind(b);
    if (fa != fb) f[fa] = fb, res += c, --cnt;
  }

  if (cnt) return 1e18;
  return res;
}

signed main()
{
  cin >> n >> m;
  for (int i = 1; i <= n; ++i) cin >> X[i], vec.push_back({n+1,i,X[i]});
  for (int i = 1; i <= n; ++i) cin >> Y[i], vec.push_back({n+2,i,Y[i]});
  for (int i = 1; i <= m; ++i) cin >> A[i] >> B[i] >> Z[i],
vec.push_back({A[i],B[i],Z[i]});
  sort(vec.begin(), vec.end());

  int res = 1e18;
  for (int i = 0; i <= 1; ++i) {
    for (int j = 0; j <= 1; ++j) res = min(res, kruskal(i,j));
  }
  cout << res << endl;
  return 0;
}
```